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ABSTRACT

Navigating safely and independently remains a persistent challenge for visually impaired individuals, particularly in unfamiliar or dynamic environments where traditional aids like white canes and guide dogs offer only limited spatial awareness and no contextual understanding of obstacles. To bridge this critical gap, we propose **VisionNav**, an intelligent, on-device mobile navigation system that integrates LiDAR based depth sensing and camera-based object recognition through sensor fusion. VisionNav utilizes the LiDAR sensor available on modern iPhones to construct a 3D real-time 3D map of the user's surroundings, accurately estimating the distance and position of nearby obstacles. Simultaneously, the RGB camera feed is processed using a lightweight YOLO26 object detection model optimized with CoreML to identify and classify objects such as doors, people, furniture, and narrow paths. Through advanced fusion of depth and visual data, the system not only detects the presence and location of obstacles but also understands their type and context, enabling highly specific navigational assistance. Real-time audio instructions and haptic feedback such as "Door ahead at 1.5 meters" or "Person on the right, move left" are delivered to guide the user safely through complex indoor and moderately challenging outdoor environments. By leveraging ARKit for scene understanding and ensuring offline, low latency operation, VisionNav provides a robust and context-aware solution that significantly enhances mobility, independence, and confidence for visually impaired users.

Keywords: *LiDAR, Sensor Fusion, Object Detection, YOLO26,, Visually Impaired, Real-time Navigation, ARKit, CoreML, iOS App, Assistive Technology, 3D Mapping, Haptic Feedback, Voice Assistance, Obstacle Detection*

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LIST OF ABBREVIATIONS

AI :	Artificial Intelligence
ARKit :	Augmented Reality Kit
CNN :	Convolutional Neural Network
CoreML :	Core Machine Learning
FPS :	Frames Per Second
iOS :	iPhone Operating System
LiDAR :	Light Detection and Ranging
mAP :	Mean Average Precision
ONNX :	Open Neural Network Exchange
OSM :	OpenStreetMap
YOLO :	You Only Look Once
AABB :	Axis-Aligned Bounding Box
dToF :	direct Time-of-Flight
FoV :	Field of View
GPS :	Global Positioning System
IMU :	Inertial Measurement Unit
IOU :	Intersection Over Union
IPO :	Input-Process-Output
KD-Tree :	K-Dimensional Tree
LiDAR:	Light Detection And swriRanging
MR.NAVI :	Mixed-Reality Navigation System
NIR :	Near-Infrared
NMS :	Non-Maximum Suppression
RANSAC :	Random Sample Consensus
RGB :	Red-Green-Blue
RoI :	Region of Interest
SDLC :	Software Development Life Cycle
SLAM :	Simultaneous Localization and Mapping
SoC :	System on Chip
SPAD :	Single-Photon Avalanche Diode

ToF : Time of Flight
UI/UX : User Interface / User Experience
VCSEL : Vertical-Cavity Surface-Emitting Laser
YOLO26 : You Only Look Once version 26

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND THEORY

Navigating the environment safely and independently remains a significant challenge for visually impaired individuals. Traditional assistive tools such as white canes and guide dogs provide only limited obstacle detection and do not give a full understanding of the surroundings. Users often rely on touch or assistance from others to move safely, which can limit their autonomy and confidence. Recent advances in computer vision, deep learning, and spatial sensing technologies have enabled the development of intelligent navigation aids that offer richer contextual information, better situational awareness, and real-time feedback. VisionNav leverages these advances by combining depth sensing, camera-based object detection, and navigation frameworks to build a robust system capable of identifying obstacles and guiding users in real time.

Depth sensing works by emitting laser pulses and measuring the time taken for the light to return after reflecting off surfaces, generating high-resolution, three-dimensional point clouds that accurately represent the spatial geometry of the environment. These point clouds are processed to create detailed maps that capture the shape, distance, and relative position of objects and surfaces around the user. Sensors like the iPhone's built-in LiDAR provide access to this spatial data through Apple's ARKit framework, enabling mobile applications to perform real-time 3D environment mapping, precise distance estimation, and recognition of environmental structures such as walls, doors, furniture, stairs, and other obstacles. This precise spatial information is especially valuable in environments where obstacles can appear suddenly, in crowded or cluttered spaces, or under poor lighting conditions such as dimly lit rooms or outdoor areas at night. By providing an accurate, continuously updated model of the surroundings, depth sensing allows the system to guide users safely and confidently, filling critical gaps left by traditional mobility aids.

Complementing depth sensing, camera-based object detection captures RGB image data and processes it using advanced deep learning models such as YOLO26. These models are trained on custom datasets specifically designed to recognize obstacles that are most relevant to visually impaired users, including doors, stairs, people, furniture, and other commonly encountered objects. By understanding both the type and context of objects, the camera enables semantic interpretation of the environment, allowing the system to differentiate between safe pathways and potential hazards. Combining this semantic understanding with depth information enhances both accuracy and safety, reducing the likelihood of collisions or confusion.

The integration of these two sensing modalities, known as sensor fusion, allows VisionNav to merge the precise geometric measurements from depth sensing with the contextual and semantic information from the camera. This fusion produces a more complete and reliable understanding of the user's surroundings. It enables the system YOLOto deliver meaningful, timely feedback through voice instructions and haptic cues, guiding users safely around obstacles and helping them navigate complex spaces confidently. Real-time processing and inference are achieved using Swift, Apple's native programming language for iOS, together with CoreML for on-device machine learning and ARKit for spatial mapping. Swift's performance, safety features, and seamless integration with Apple's ecosystem ensure the application runs efficiently and responsively on mobile devices, making it practical for everyday use.

By leveraging modern hardware, trained machine learning models, and intelligent software frameworks, VisionNav aims to provide visually impaired users with enhanced mobility, independence, and confidence in navigating both indoor and moderately complex outdoor environments. Its design emphasizes real-world usability, combining accuracy, speed, and context awareness to make navigation safer and more intuitive than traditional mobility aids or single-sensor systems.

1.2 PROBLEM STATEMENT

Visually impaired individuals face significant challenges when navigating unfamiliar, cluttered, or changing environments due to limited access to spatial and contextual information. Traditional mobility aids, such as white canes and guide dogs, can detect nearby obstacles but provide little information about their type, position, or behavior. As a result, users often rely on physical contact or human guidance, which limits independence and increases the risk of collisions or disorientation.

Recent technological advances have introduced assistive navigation systems using either camera-based object detection or depth-sensing devices like ultrasonic sensors and LiDAR. Camera-based systems can recognize and classify objects but often struggle to estimate distances accurately, especially in low-light or high-glare conditions. Depth-sensing technologies provide precise spatial measurements but cannot identify the nature of obstacles — they detect that “something” is present but not what it is. Operating these systems in isolation leads to incomplete or ambiguous feedback, reducing trust and effectiveness in real-world scenarios.

Additionally, many existing solutions are not optimized for mobile platforms or require external hardware, making them impractical for continuous, real-time use. There is a clear need for a **compact, mobile, and intelligent navigation system** that integrates both precise spatial awareness and semantic understanding to provide meaningful and actionable feedback.

1.3 OBJECTIVE

The primary objectives of this project are:

- To develop a real time mobile navigation system with spatial and obstacle awareness.
- To provide audio and haptic feedback for safe, independent navigation offline.

1.4 SCOPE AND APPLICATION

1.4.1 SCOPE

This project develops a mobile navigation assistant for visually impaired users, focusing on short-range obstacle detection and guidance. The system uses sensor fusion by combining LiDAR depth sensing and camera-based object detection, and is implemented on iOS devices with LiDAR sensors for reliable performance.

The project includes the following features:

- Real-time detection and classification of obstacles such as walls, doors etc.
- Accurate distance estimation using LiDAR for precise spatial awareness.
- Offline operation, with no internet or cloud dependency.
- Voice alerts and haptic feedback for real-time guidance.

Out of Scope:

- Navigation beyond 2 meters.
- Navigation requiring GPS or large-scale path finding.
- Advanced mapping techniques like full SLAM.
- Multilingual voice feedback or customization.

1.4.2 APPLICATION

VisionNav enhances the mobility and independence of visually impaired users by providing real-time obstacle detection, object classification, and immediate audio-haptic feedback.

Key applications include:

- **Assistive Mobility Tool:** Real-time alerts for doors, furniture, people, and pathways using voice and vibration feedback.
- **Indoor Navigation:** Supports safe movement in homes, offices, schools, and public buildings.
- **Public Facility Navigation:** Helps detect objects and hazards in malls, hospitals, airports, and other complex spaces.
- **Future Use in Robotics and Smart Devices:** Sensor fusion framework can be adapted for autonomous navigation.
- **Educational and Research Tool:** Foundation for research in sensor fusion, augmented reality, mobile machine learning, and accessibility solutions.

CHAPTER 2: LITERATURE REVIEW

Research in recent years has explored various ways to help visually impaired individuals navigate independently using technology. This review examines studies in four areas relevant to VisionNav: Combining sensor data for better perception, Efficient object detection for mobile devices, Existing navigation aids, and Use of depth sensing and LiDAR in accessibility tools.

2.1 Sensor Fusion for Context-Aware Navigation

Early navigation systems often used single sensors, which had limitations. For example, the vOICe system [6] converted images to sound, but could not measure distance or recognize specific objects. More recent research shows that combining multiple sensors produces better results. In one study, Pradeep et al. [9] combined stereo camera data with motion sensor data in a wearable device, which reduced false obstacle detections by 40% compared to camera-only systems. Similarly, Zhang and Chen [13] used deep learning to combine LiDAR and camera data, improving both object recognition and distance measurement. Their work shows that sensor fusion is necessary to create detailed environmental maps. Lee et al. [5] applied this approach to mobile devices, showing that iPhones with LiDAR and ARKit can measure distance accurately within 0.1m at 3 meters. This level of accuracy is essential for safe navigation systems like VisionNav.

2.2 Efficient Object Detection for Mobile and Real-Time Applications

Object detection on mobile devices must be both fast and accurate. The YOLO (You Only Look Once) system, introduced by Redmon et al. [10], made real-time object detection possible. Since then, researchers have worked to make YOLO lighter and faster for mobile use. Wang et al. [12] designed a simpler neural network that runs quickly on phones while still detecting objects accurately. For VisionNav, we used work by Jocher et al. [3] to prepare our YOLO model for Apple's CoreML system, which helps us achieve the speed needed for real-time feedback.

2.3 Assistive Navigation Systems for the Visually Impaired

Several existing applications help visually impaired users. Microsoft's Seeing AI and Google Lookout use smartphone cameras to read text, identify objects, and describe scenes. While helpful, these apps focus more on identifying objects than guiding navigation, and they do not provide precise distance information [4]. Research systems have tried to address this. Gupta et al. [2] created a wearable device called "Sound of Vision" that uses depth cameras to create sound-based maps of surroundings.. Pfister et al. [8] developed a mixed-reality system called MR.NAVI that builds maps and gives spoken instructions, showing the value of clear verbal guidance. These studies reveal a need for a system that combines depth sensing, object recognition, and real-time guidance in a single mobile device, which is what VisionNav aims to provide.

2.4 Depth Sensing and LiDAR in Accessibility Applications

The recent availability of LiDAR sensors in consumer phones has created new opportunities for navigation aids. In robotics, Murai et al. [7] used 360-degree LiDAR to help robots navigate indoors, proving that detailed 3D maps are useful for obstacle avoidance. For human use, Smith and Johnson [11] tested the iPhone's LiDAR for detecting stairs and overhead obstacles. Their work describes how to process LiDAR data, from capturing it with ARKit to filtering out floors using algorithms like RANSAC. Apple's documentation on ARKit [1] also explains how LiDAR, camera, and motion data work together, providing the foundation for building systems like VisionNav.

2.5 Synthesis and Identification of the Research Gap

The studies reviewed show that effective navigation aids need to combine different types of data. However, most current systems have limitations: some need extra hardware, others rely on cloud processing, and many do not provide both accurate distance measurement and object recognition in real time. There is a clear need for a self-contained, mobile system that works offline, protects privacy, and gives immediate, useful guidance. VisionNav addresses this gap by using the iPhone's built-in sensors and processing all data on the device.

CHAPTER 3: METHODOLOGY

3.1 PROCESS MODEL

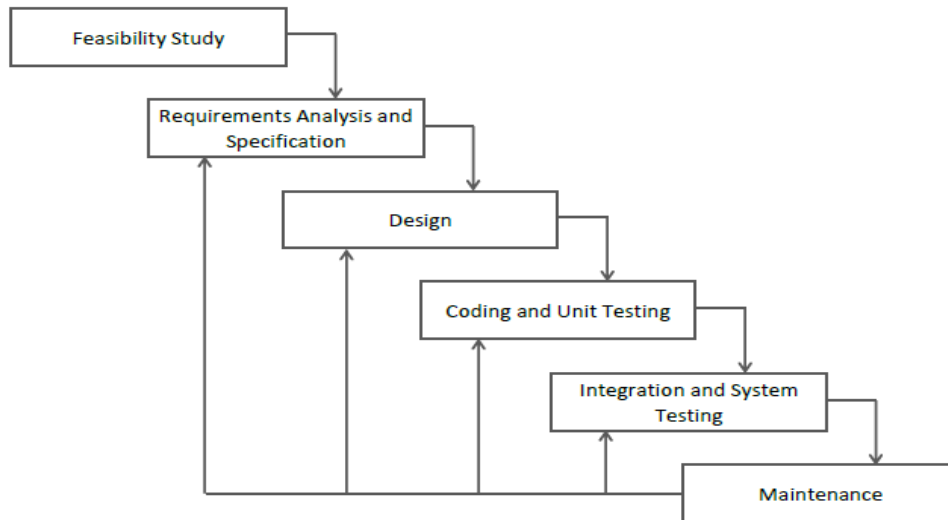


Figure 1: Iterative Waterfall Model

The Iterative Waterfall Model used for VisionNav is a refined version of the traditional waterfall software development lifecycle (SDLC) that incorporates iteration and feedback loops between phases. Unlike the strictly linear approach, this model allows revisiting earlier stages after each iteration, enabling gradual refinement and better handling changes or unforeseen issues. This approach suited VisionNav well, as integrating sensor fusion, real-time machine learning, and mobile application development required iterative testing, validation, and adjustments.

The project proceeded through sequential phases, and after completing each phase, results were reviewed and refined in successive iterations. This ensured that design decisions and implementations improved continually based on testing and feedback. The phases of the Iterative Waterfall Model and progress so far are summarized below:

Phase 1: Feasibility Study

In the initial phase, we assessed the viability of VisionNav across multiple dimensions:

- **Technical Feasibility:** Verified that iOS devices equipped with LiDAR and cameras (using ARKit and CoreML) could support real-time sensor fusion, object detection, and haptic/audio feedback integration.
- **Financial Feasibility:** Estimated development, testing, and deployment costs. Leveraging existing iOS devices and on device computation minimized recurring expenses.
- **Operational Feasibility:** Evaluated practical deployment scenarios, including indoor navigation, limited outdoor use, and offline operation.
- **Legal Feasibility:** Reviewed data protection and accessibility compliance all processing occurs on device, minimizing privacy risks.

This phase confirmed that VisionNav was technically achievable, economically viable, operationally practical, and legally compliant.

Phase 2: Requirement Analysis

During this phase, requirements were defined and documented:

- **Functional Requirements:** Real-time detection of obstacles such as doors, people, and furniture; semantic labeling; haptic and audio feedback.
- **Non-functional Requirements:** Real-time performance (≥ 15 fps), reliable offline operation, battery efficiency, and accessible user interface.
- **User Needs:** Close-range (≤ 2 m) obstacle awareness with intuitive feedback for visually impaired users.
- **Constraints:** Swift, ARKit, and CoreML on iOS devices, with a 2 m detection range limit.

Phase 3: System Design

Based on requirements, the system architecture was developed:

- Designed the sensor fusion pipeline integrating LiDAR depth data with YOLO26-based camera detection.

- Defined data flow from sensors to machine learning inference, decision logic, and user feedback outputs.
- Refined the design iteratively to optimize performance on iOS devices and ensure smooth user experience.

Phase 4: Implementation (Coding)

During implementation:

- VisionNav was developed in Swift, leveraging ARKit for sensor data and CoreML for on-device inference with YOLO26.
- Code was written and tested module by module, allowing early debugging and iterative refinements.

Phase 5: Testing

Testing was performed to ensure system reliability and performance:

- **Unit Testing:** Validated individual modules (sensor capture, detection, fusion, feedback).
- **Integration Testing:** Verified end-to-end interaction (e.g., detecting a door and issuing feedback).
- **Performance Testing:** Checked inference speed, latency, and battery consumption.

Phase 6: Deployment

After testing:

- The app was packaged for **iOS devices** (iPhone 12 Pro or newer).
- Deployment adhered to App Store guidelines or institutional distribution policies.

Phase 7: Maintenance

Post-deployment activities included:

- Periodic updates for bug fixes, performance optimizations, and enhancements.
- Monitoring compatibility with new iOS updates and hardware upgrades (e.g., enhanced LiDAR in newer devices).

3.2 SYSTEM BLOCK DIAGRAM:

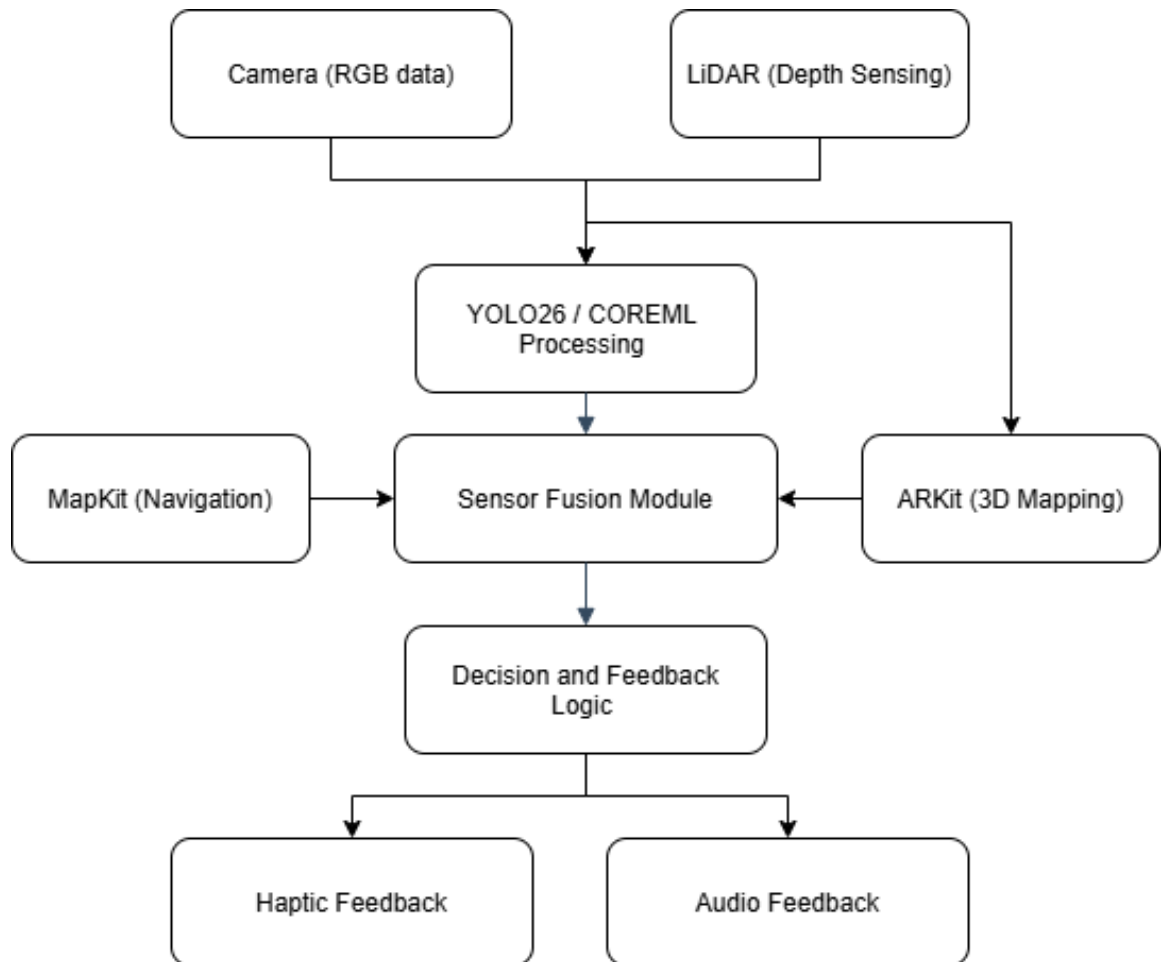


Figure 2: System Block Diagram

The above diagram illustrates the VisionNav system architecture, structured according to the Input-Process-Output (IPO) framework.

3.2.1 Input

The VisionNav system receives data from two primary sensors. The **LiDAR sensor** emits laser pulses to generate high-resolution 3D point clouds, providing precise spatial geometry of the environment. The **RGB camera** captures visual images that contain semantic information about objects such as doors, stairs, people, and furniture. Together, these inputs provide both spatial and contextual information necessary for effective navigation.

3.2.2 Processing

The system processes the raw sensor data in several stages:

1. **Preprocessing:** LiDAR data is filtered to remove noise, down sampled to reduce computational load, and aligned for stable 3D mapping. Camera images are resized, normalized, and optionally augmented for better ML inference.
2. **Sensor Fusion:** Preprocessed LiDAR and camera data are combined, aligning 3D point clouds with corresponding image pixels to produce a coherent fused 3D representation of the surroundings.
3. **Machine Learning Inference:** The fused data is analyzed by CoreML running YOLO26 models trained on custom datasets. Objects are detected, classified, and their distances estimated in real time.
4. **Decision and Feedback Logic:** Obstacle priority is determined based on distance and relevance. Instructions are generated for the feedback layer to guide the user safely.

3.2.3 Output

The system communicates the processed information to the user through:

Audio alerts: Voice prompts indicating obstacle type and distance.

Haptic feedback: Vibration patterns corresponding to proximity and urgency.

This structured workflow ensures that VisionNav converts complex sensor data into actionable real time guidance for object detection and navigation purposes.

3.4 ALGORITHMS

3.4.1 YOLO:

YOLO (You Only Look Once) is a modern single stage object detection algorithm built on deep Convolutional Neural Network (CNN) principles and optimized for real-time performance. Unlike two-stage detectors that first generate region proposals and then classify them, YOLO performs localization and classification in a single forward pass of the network. Its architecture is composed of three main components: a backbone, which extracts hierarchical visual features from the input image using convolutional layers; a neck, which fuses multi-scale features to enable detection of objects of different sizes; and a detection head, which predicts bounding box coordinates, objectness confidence scores, and class probabilities at each spatial location

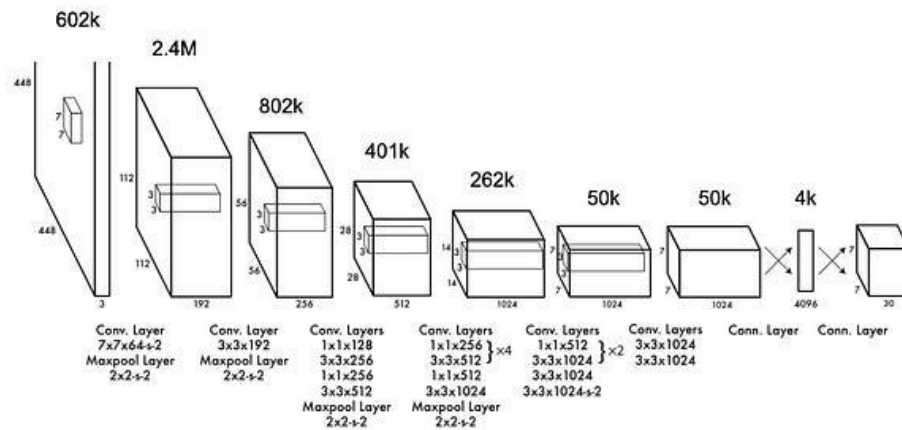


Figure 3: YOLO Architecture.

The above figure illustrates the CNN architecture used in the YOLO detection model. The input image is processed through a sequence of convolutional and max-pooling layers to extract hierarchical features. The numbers written above each block (e.g., 602k, 2.4M, 802k) indicate the approximate number of trainable parameters in each group of layers, showing how model complexity increases with depth. The values below each block (such as 64, 192, 256, 512, and 1024) represent the number of feature channels produced by the corresponding convolutional layers. As the network goes deeper, spatial resolution decreases while semantic richness increases, enabling accurate object recognition.

The YOLO algorithm treats object detection as a single regression problem, straight from image pixels to bounding box coordinates and class probabilities. The process is defined by four primary mathematical and structural approaches.

1. Residual blocks

The algorithm begins by dividing the input image into an $N \times N$ grid. Each individual grid cell is responsible for detecting an object if the center of that object falls within the bounds of that cell. Each cell predicts a fixed number of bounding boxes and a confidence score for those boxes. This confidence score reflects how certain the model is that the box contains an object and how accurate it thinks the predicted box is.

2. Bounding box regression

The model uses a single regression module to predict the attributes of objects. For every bounding box, the algorithm generates a vector Y that represents the state of the detection:

$$Y = [p_c, b_x, b_y, b_h, b_w, c_1, c_2]$$

Where:

- p_c : The probability score (confidence) that an object exists within the grid cell.
- b_x, b_y : The coordinates of the center of the bounding box relative to the boundaries of the grid cell (normalized between 0 and 1).
- b_h, b_w : The height and width of the bounding box relative to the entire image dimensions.
- $c_1, c_2, \dots, c_n$: The one-hot encoded class probabilities (e.g., "Door," "Stair," or "Person").

3. Intersection Over Unions or IOU

The goal of the IOU (a value between 0 and 1) is to discard such grid boxes to only keep those that are relevant. The user defines its IOU selection threshold, which can be, for instance, 0.5.

Then, YOLO computes the IOU of each grid cell, which is the Intersection area divided by the Union Area. Finally, it ignores the prediction of the grid cells having an $\text{IOU} \leq \text{threshold}$ and considers those with an $\text{IOU} > \text{threshold}$.

4. Non-Max Suppression or NMS

While IoU filters out poor candidates, an object may still have multiple high-scoring boxes that passed the IoU threshold. Non-Maximum Suppression (NMS) is the final refinement step used to remove redundant detections.

- **YOLO26:**

Unlike previous versions YOLO26, the latest variant of YOLO utilizes an NMS-free (Non-Maximum Suppression) design and has removed Distribution Focal Loss (DFL), which significantly slashes the latency between the iPhone's camera frame capture and the object detection output.

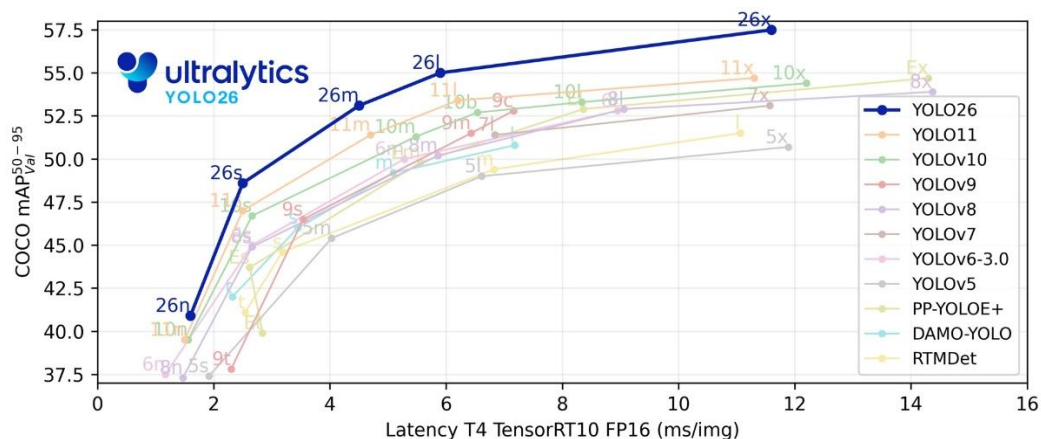


Figure 4: Performance benchmarks of YOLO26 relative to other architectures.

This diagram compares YOLO object detection models by showing the trade-off between **speed and accuracy**. The x-axis is inference latency (ms per image on an NVIDIA T4 with TensorRT FP16), and the y-axis is COCO mAP (detection accuracy). Each curve is a YOLO version with different model sizes (n, s, m, l, x). It shows that as models get bigger, accuracy improves but latency increases. The blue YOLO26 line sits on the top edge, meaning it **gives** higher accuracy at the same or lower latency than earlier YOLO versions.

This streamlined architecture means that when VisionNav processes environmental objects like doors or furniture, the model produces cleaner, end-to-end predictions without the typical post-processing bottleneck. Because YOLO26's structure is simplified and optimized for mobile hardware, the resulting CoreML file runs with a significantly smaller memory footprint. This efficiency allows VisionNav to generate audio and haptic feedback with millisecond precision, ensuring that the spatial understanding provided by YOLO is not just accurate, but fast enough to maintain user safety in dynamic, high-stakes environments.

3.4.2 Depth and Obstacle Mapping Algorithm:

The VisionNav system utilizes a high-frequency LiDAR (Light Detection and Ranging) pipeline to transform unstructured environmental data into a semantically labeled 3D map. To ensure real-time performance on a mobile SoC (System on Chip), the following algorithmic stages are implemented:

Stage 1: Spatial Pre-processing and Voxelization

LiDAR scans generate high-density point clouds containing $P = \{p_1, p_2, \dots, p_n\}$ points, where each p_i is a vector (x, y, z) . To reduce computational latency, the algorithm applies:

- **Voxel Grid Down sampling:** The 3D space is partitioned into a grid of voxels with dimensions $v_{size} = (v_x, v_y, v_z)$. For each voxel, a centroid $p_{centroid}$ is calculated to represent all internal points, reducing the point count by up to 90% while preserving spatial topology.
- **RoI Truncation:** A visibility frustum is defined to crop points outside the functional navigation range. Points are filtered based on the constraint:

$$Z_{min} \leq z \leq Z_{max} \text{ and } |x| \leq X_{threshold}$$

Typically, for indoor navigation, Z_{max} is capped at 5.0 meters to prioritize immediate obstacles.

Stage 2: Ground Plane Segmentation (RANSAC)

To isolate potential obstacles, the dominant horizontal plane (the floor) must be identified and removed. This is achieved through the Random Sample Consensus (RANSAC) algorithm:

1. **Model Definition:** The algorithm assumes a plane equation $ax+by+cz+d=0$.
2. **Iterative Estimation:** It randomly samples three points to compute a candidate plane.
3. **Inlier Scoring:** For each iteration k , the distance D of all points to the plane is calculated.

A point is an inlier if:

$$D = \frac{|ax_i + by_i + cz_i + d|}{\sqrt{a^2 + b^2 + c^2}} < \epsilon$$

Where ϵ is the distance threshold (e.g., 0.05m). The points identified as the ground plane are subtracted from the cloud, leaving a set of "Outlier" points representing obstacles.

Stage 3: Spatial Clustering and Object Extraction

The remaining non-ground points are grouped into discrete objects using **Euclidean Clustering** optimized with a **KD-Tree** spatial index:

- **Metric Search:** For a point p_i , the algorithm searches for neighbors p_j within a distance d_{thresh} .
- **Connectivity Logic:** If $|p_i - p_j| < d_{thresh}$, the points are assigned to the same cluster c_k .
- **Density Constraints:** Clusters with a point count $n < d_{thresh}$ are discarded as sensor noise, ensuring that only significant physical structures (e.g., furniture, pedestrians) are processed.

Stage 4: Volumetric Bounding and Distance Estimation

Each extracted cluster c_k is encapsulated within an **Axis-Aligned Bounding Box (AABB)** defined by the extrema:

$$Box_k = \{(x_{min}, x_{max}), (y_{min}, y_{max}), (z_{min}, z_{max})\}$$

The absolute distance d to the obstacle is derived from the LiDAR's Time-of-Flight (ToF) principle:

$$d = \frac{c \cdot \Delta t}{2}$$

Where c is the speed of light ($\approx 3 \times 10^8$ m/s) and Δt is the round-trip time. This numerical data is synchronized with the RGB camera feed for sensor fusion, allowing the system to output both the identity and the precise 3D coordinate of the obstacle to the user.

3.4.3. Navigation Algorithm

The navigation algorithm of VisionNav is based on graph-based shortest-path planning integrated with MapKit. The environment is represented as a weighted graph ($G = (V, E)$), where nodes ($v \in V$) correspond to significant points such as intersections or landmarks, and edges ($e \in E$) represent paths connecting nodes, weighted by distance:

$$w(e_{ij}) = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$$

Where (x_i, y_i) and (x_j, y_j) are the coordinates of connected nodes v_i and v_j

MapKit's MKDirections API is used to compute the optimal route on this graph. Internally, MapKit performs a shortest-path computation similar to Dijkstra's algorithm, generating a sequence of route points (waypoints) from the user's current location to the destination. Formally, given start node (v_{start}) and goal node (v_{goal}), the shortest path is:

$$\text{Path} = \arg \min_{\text{paths } P: v_{start} \rightarrow v_{goal}} \sum_{e \in P} w(e)$$

Once the path is computed, it is converted into ordered waypoints (W_i). The user's position is continuously tracked using CoreLocation, and the distance (D_i) and heading (θ_i) to the next waypoint are calculated as:

$$D_i = \sqrt{(x_{curr} - x_{W_i})^2 + (y_{curr} - y_{W_i})^2} \quad \theta_i = \text{atan2}(y_{W_i} - y_{curr}, x_{W_i} - x_{curr})$$

At each step during navigation, the next waypoint W_i is selected, and instructions are issued (e.g., "Head towards W_i " or "Turn to heading θ_i ") until the destination is reached. MapKit handles the underlying routing, map representation, and path updates, allowing the algorithm to focus on navigation logic.

3.4 FLOWCHART:

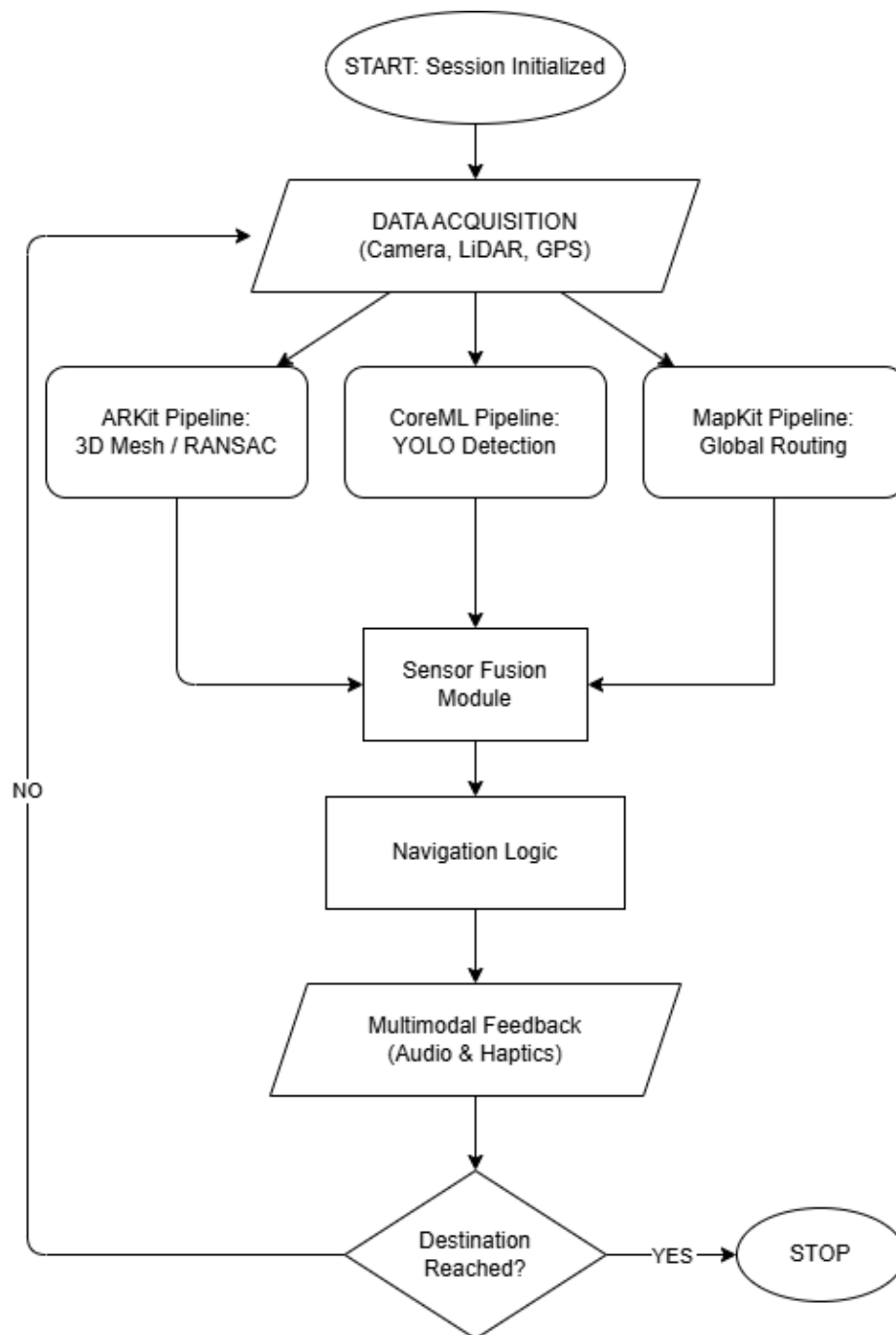


Figure 5: Flowchart for the VisionNav Obstacle Detection and Navigation System

3.5 UML DIAGRAMS:

3.5.1 Sequence Diagram

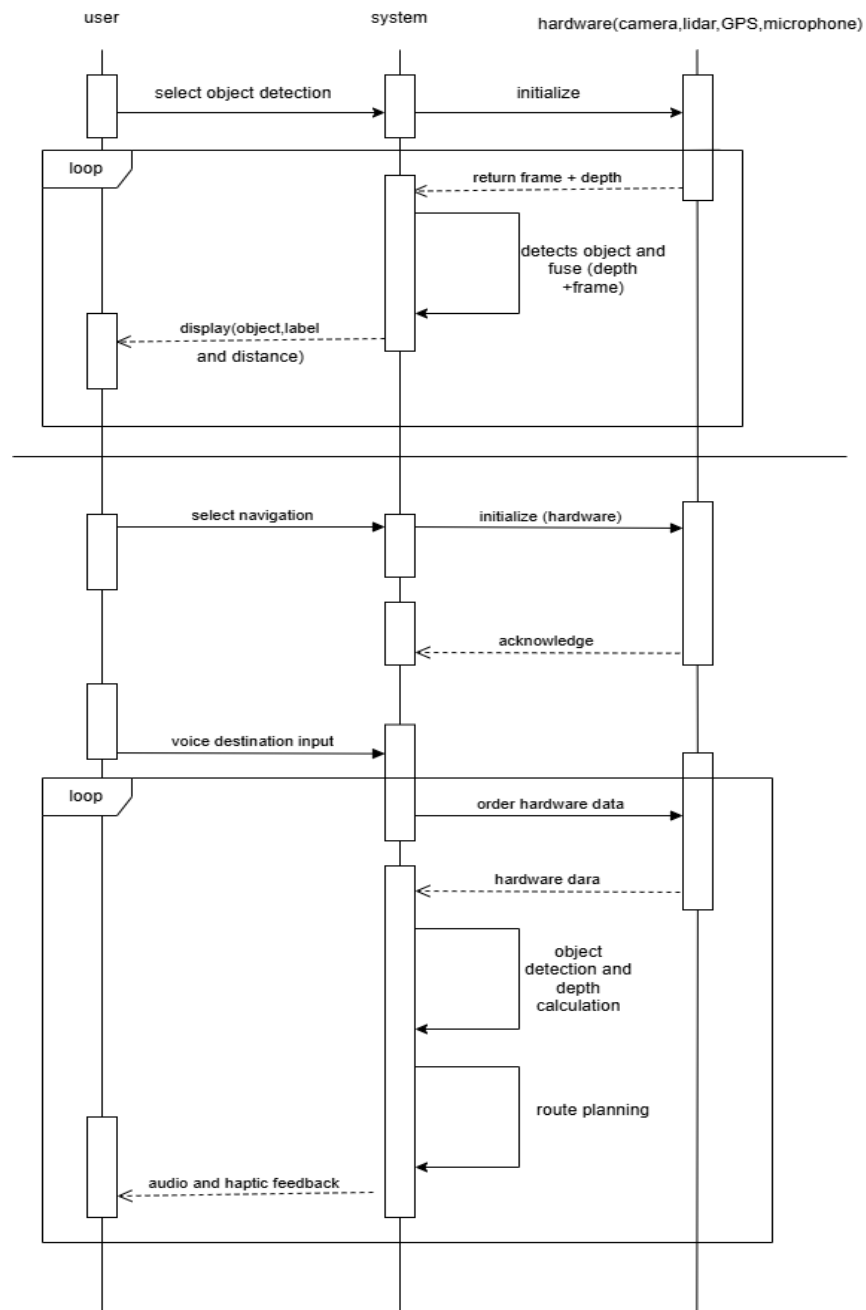


Figure 6: Sequence diagram for VisionNav

Sequence Diagram Description:

The **Object Detection** sequence in VisionNav begins when the user selects the Object Detection option in the app. The system then initializes the required hardware components, including the iPhone RGB camera and LiDAR sensor. Once initialized, the hardware captures 1080p camera frames and depth data, which are sent back to the system. The system continuously performs object detection, fusing depth information with the captured image frames. Detected objects, along with their distances from the user, are displayed in real-time, and this process runs in a continuous loop to ensure up-to-date detection of the surroundings.

For the **Navigation** sequence, the user first selects the Navigation option in the app, prompting the system to initialize all necessary hardware components. The hardware acknowledges its readiness, after which the user provides the destination using voice input. The system then repeatedly requests live sensor data, including camera and LiDAR information. Using this data, the system performs object detection and depth calculations, which feed into route planning to navigate safely around obstacles. Finally, the system provides audio and haptic feedback to guide the user along the chosen path, ensuring real-time and intuitive navigation for visually impaired users.

3.5.2 Use Case Diagram

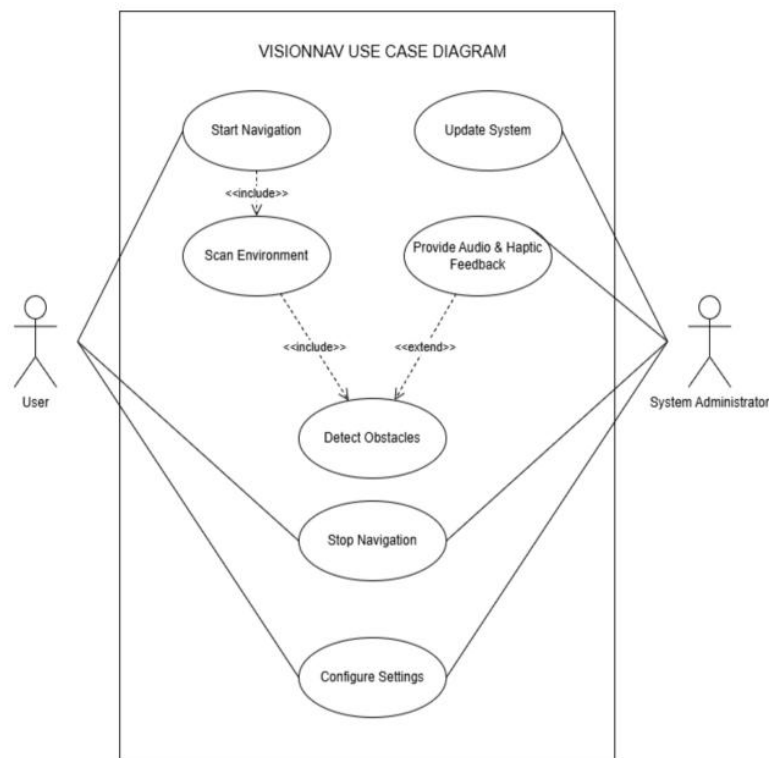


Figure 7: Use Case Diagram of the VisionNav Application

Actors and Their Relations:

i. User

- Start Navigation
- Scan Environment
- Detect Obstacles
- Stop Navigation
- Configure Settings

ii. System Administrator

- Update System
- Provide Audio & Haptic Feedback
- Stop Navigation
- Configure Settings

3.6 TOOLS USED

3.6.1 Software and Frameworks:

1. Swift

Swift is a programming language developed by Apple for iOS and macOS development. VisionNav uses Swift as the main language to handle all parts of the app, including processing sensor data, running machine learning models, updating the user interface, and managing navigation logic. SwiftUI is a framework that allows building interfaces in a declarative way. It makes it easy to update the interface automatically when the user moves or when sensor data changes, which helps VisionNav respond quickly to real-time events.

2. Xcode

Xcode is the official development environment for iOS apps. It allows writing, testing, and debugging code, as well as running the app on simulators or real devices. VisionNav uses Xcode to integrate different frameworks like ARKit, CoreML, and MapKit, and to make sure the app runs smoothly on iPhones.

3. ARKit

ARKit is a framework for augmented reality. It combines the camera view and motion sensors to track the device's position and orientation. In VisionNav, ARKit helps the app understand the surrounding environment, know where the user is, and place virtual waypoints on the screen to guide navigation.

4. AVFoundation (Camera)

AVFoundation provides control over the iPhone camera. It allows capturing video with specific resolution, frame rate, and focus. VisionNav uses AVFoundation to get a live video feed for processing by the object detection system. This ensures that the app can detect objects in real time. AVFoundation also provides text-to-speech features. VisionNav uses this to give clear voice instructions, such as telling the user when to

turn or alerting them about upcoming landmarks. This makes navigation hands-free and suitable for visually impaired users.

5. CoreML

CoreML is a framework for running machine learning models on the device. VisionNav uses CoreML to detect objects in the camera feed. Running the models on the phone reduces delay and protects the user's privacy. It also allows the app to respond quickly to the surroundings.

6. MapKit

MapKit allows adding interactive maps, finding addresses, and planning routes inside iOS apps. VisionNav uses MapKit to show the user's location on a map, calculate walking paths, and provide turn-by-turn directions. It helps the user stay aware of their position while navigating.

7. CoreHaptics

CorePHaptics is Apple's framework for creating precise, customizable tactile feedback on iOS devices. It allows developers to design vibrations with varying intensity, sharpness, and patterns, enabling users to perceive information through touch. The framework supports real-time haptic responses, making interactions more intuitive and immersive, and can be combined with audio cues for enhanced sensory feedback. This technology is used in applications like VisionNav to translate environmental data into tactile signals for navigation.

3.6.2 Hardware Tools

1. LiDAR Sensor

The VisionNav spatial awareness engine is centered on a Solid-State direct Time-of-Flight (dToF) LiDAR sensor. Unlike traditional mechanical LiDAR systems that utilize rotating mirror assemblies, the mobile-integrated dToF sensor employs a fixed Vertical-Cavity Surface-Emitting Laser (VCSEL) array, making it ideal for the power-constrained environments of handheld assistive technology.

Operational Principle (Time-of-Flight):

The primary metric of the LiDAR sensor is the precise measurement of the temporal interval between signal emission and backscatter detection. The sensor operates in the Near-Infrared (NIR) spectrum, typically at a wavelength of 940 nm.

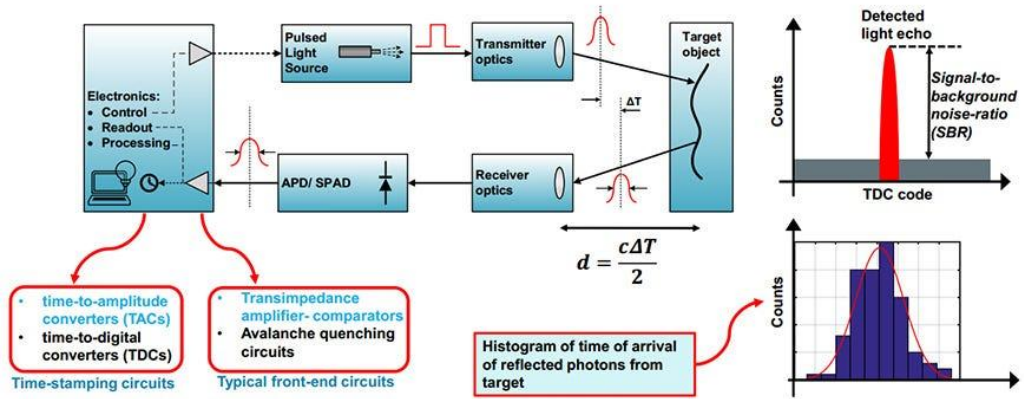


Figure 8: Time of flight Working Principle

For each individual pulse in the emitted grid, the distance d to a reflecting surface is determined by:

$$d = \frac{c \cdot \Delta t}{2}$$

Where, c is the speed of light in the medium ($\approx 3 \times 10^8 \text{ m/s}$) and Δt is the measured round-trip time-of-flight.

Spatial Data Acquisition and Processing

The hardware architecture consists of a high-sensitivity Single-Photon Avalanche Diode (SPAD) receiver array. This allows the system to detect individual photons, ensuring high precision even in low-light environments.

- 1. **Point Cloud Generation:** The sensor projects a non-uniform grid of approximately 570–600 points. These points are captured at frequencies up to 60 Hz, providing a high-temporal-resolution 3D point cloud of the forward field of view (FoV).
- 2. **Scene Reconstruction (Meshing):** Through the ARKit framework, the unstructured point cloud is processed to a Triangular Mesh. This meshing process involves the integration of depth data with Simultaneous Localization and Mapping (SLAM) data from the IMU (Inertial Measurement Unit).
- 3. **Semantic Classification:** The A-series Neural Engine performs real-time semantic segmentation on the mesh, identifying topological features such as:
 - Planar Surfaces: Floors and walls.
 - Non-Planar Clusters: Furniture, pedestrians, and structural obstacles.

Technical Specifications of iPhone LiDAR sensor:

The following table summarizes the performance benchmarks of the mobile LiDAR integrated into the VisionNav prototype:

Specification	Parameter Value
Sensor Type	Solid-State dToF (VCSEL/SPAD)
Effective Range	0.5 m to 5.0 m
Sampling Frequency	15–60 Hz
Depth Precision	±1 cm at 2 m range
Operating Wavelength	940 nm (Near-Infrared)
Field of View (FoV)	≈ 60° Horizontal × 45° Vertical

In the VisionNav pipeline, LiDAR data acts as the high-precision Spatial Ground Truth. While the RGB camera (via YOLO) provides semantic "What" labels (e.g., Chair), the LiDAR sensor provides the "Where" (e.g., "2.34 meters at 15 degree azimuth"). This Sensor Fusion ensures that navigational cues are physically grounded, allowing the system to distinguish between an image of an object and a tangible physical obstacle.

2. RGB Camera:

The iPhone RGB camera captures high-resolution color images using a Red-Green-Blue (RGB) sensor array.

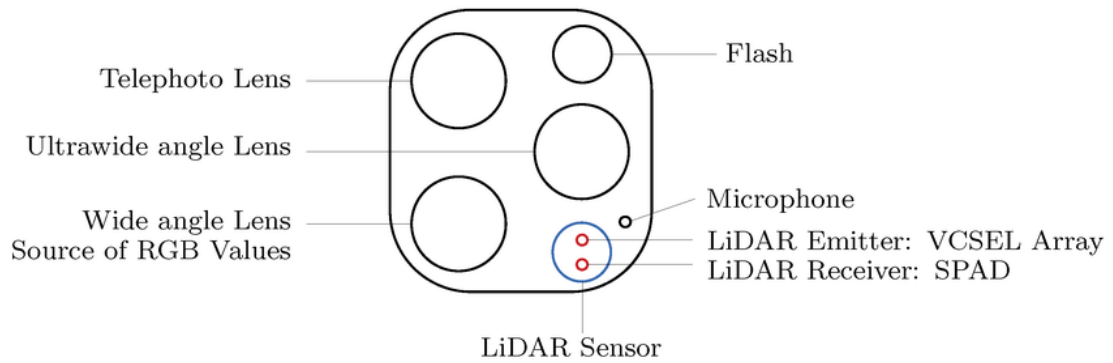


Figure 9: iPhone Camera Cluster

It records visual information by detecting light intensity in the three primary color channels, enabling accurate color reproduction and detailed imagery. Combined with advanced image processing, the camera supports features like object recognition, depth estimation (when fused with LiDAR), and real-time visual analysis. In VisionNav, the RGB camera provides essential visual context to identify obstacles, landmarks, and pathways for visually impaired users.

CHAPTER 4: EPILOGUE

4.1 TASK COMPLETED

The project has successfully transitioned from the conceptual phase to a functional prototype. The following milestones have been achieved:

- **System Architecture & Design:** The fundamental application layout and structural flow are defined, establishing a robust framework for sensor integration.
- **UI/UX Development:** The primary user interface has been developed using **Swift**, focusing on a clean and accessible layout.
- **Hardware Deployment:** The application has been successfully compiled and deployed on physical iOS hardware (LiDAR-enabled devices), ensuring real-time functionality.
- **Spatial Framework Implementation:** Initial integration of **ARKit** and **CoreML** has been completed, enabling the application to access the neural engine and depth sensors.
- **Depth Sensing Prototype:** Basic acquisition of **LiDAR depth data** has been achieved, allowing for rudimentary distance estimation.
- **Object Detection Baseline:** A functional prototype for computer-vision-based object detection has been implemented and validated against a set of common indoor obstacles.

4.2 TASK REMAINING

To reach a production-ready state, the following technical objectives are remaining to be addressed:

- **Spatial Refinement:** Enhancement of LiDAR-based obstacle mapping algorithms to improve spatial resolution and reduce false-positive detections.
- **Optimization of Vision Pipeline:** Full synchronization and latency optimization of the **YOLO/CoreML** object detection model with the live RGB camera feed.

- **Performance Engineering:** Tuning the inference engine to improve frame-per-second (FPS) performance for smoother real-time classification.
- **Advanced Navigation Logic:** Implementation of indoor/semi-outdoor path finding that dynamically updates based on real-time LiDAR obstacles.
- **Systemic Integration:** Merging global navigation features with local obstacle avoidance data for a unified "Sensor Fusion" experience.
- **Validation & Field Testing:** Extensive field testing in diverse environments (e.g., shopping malls, sidewalks) followed by final performance optimization.

4.3 GANTT CHART

	5 th Semester						6 th Semester				
Task	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
Documentation											
Feasibility Study											
Requirement Analysis											
System Design											
Implementation											
Testing											
Deployment											
Maintenance											

Figure 10: Gantt Chart

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SCREENSHOTS

